

First-Principles Study of Lithium Adsorption and Diffusion on Graphene with Point Defects

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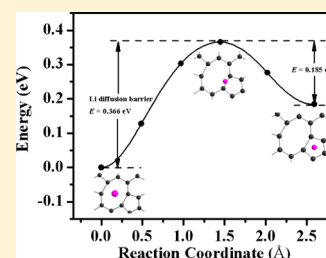
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S Supporting Information

ABSTRACT: To understand the effect of point defects on the Li adsorption on graphene, we have studied the adsorption and diffusion of lithium on graphene with divacancy and Stone–Wales defect using the first-principles calculations. Our results show that in the presence of divacancy Li adatom energetically prefers the hollow site above the center of an octagonal ring rather than the top sites of carbon atoms next to vacancy site. In the case of Stone–Wales defect, Li atom is energetically favorable to be adsorbed on the top site of carbon atom in a pentagonal ring shared with two hexagonal rings, and such adsorption results in a bucking of graphene sheet. For divacancy and Stone–Wales defects in graphene, their interactions with a Li adatom are attractive, suggesting that the presence of point defects would enhance the Li adsorption on graphene. The difference charge density and the Bader charge analysis both show that there is a significant charge transfer from Li adatom to its nearest neighbor carbon atoms.



1. INTRODUCTION

With the development of state-of-the-art energy technologies, the rechargeable lithium ion batteries (LIBs) are currently the predominant power sources for portable electronic vehicles, advanced electronics, and miscellaneous power devices.^{1,2} The main challenge in this field is to search new materials with the higher charge capacity and better cycling performance for LIB. Graphite has been widely used as anode material in LIB due to the fact that carbon in graphite can feasibly intercalate with Li. However, the capacity limit of 3D graphite is only 372 mAh/g,³ leading to its limited performance in LIB. Recent experimental studies demonstrated that lowering the dimensionality of conventional anode materials via nanotechnology can achieve higher capacity. For instance, materials with low dimensionality, such as fullerene,^{4,5} carbon nanotube,^{6–10} silicon nanowires,^{11,12} and graphene^{13–20} have been intensively investigated for the possibility to replace graphite as anode in LIBs. Among them, graphene has attracted immense attention because of its 2D crystal structure with atomic thickness, unique electronic structure, high intrinsic mechanical strength, high thermal conductivity, and large surface area.^{21,22}

During the growth process of graphene, vacancies may be formed in low concentrations. Alternatively, vacancies may be created intentionally by irradiating materials with electrons or ions or by chemical treatments.²³ Recently, several structural defects in graphene were observed at an atomic resolution by the transmission electron microscope (TEM)^{24–26} and the scanning tunneling microscope (STM).^{27–29} As predicted by recent density functional theory (DFT) calculations, some structural defects in graphene can induce localized levels close to the Fermi level (E_F), leading to local charging³⁰ or local

magnetic moments.³¹ Therefore, structural defects would have a strong influence on the electronic, optical, thermal, and mechanical properties of graphene.²³ A recent study by DFT calculations predicted that the presence of edges in graphene affects not only the reactivity of the carbon material toward the adsorption of Li adatoms but also their diffusion properties.³² It raises a question whether the defect-induced localized states also have significant impacts on the Li adsorption and diffusion on graphene. With this motivation, we have carried out the first-principles calculations based on DFT to study the lithium adsorption and diffusion on graphene with point defects. In the present study, we consider divacancy and Stone–Wales defects as representatives of point defects in graphene to study the structural and energetics aspects of Li adsorption on defective graphene. Meanwhile, we analyze the charge transfer between Li adatom and its surrounding C atoms. The energy barriers for Li migration on the basal plane of graphene with and without point defects are evaluated.

2. THEORETICAL CALCULATIONS

All calculations are performed by using the projector-augmented wave (PAW) method,³³ as implemented in the Vienna ab initio simulation package (VASP).³⁴ The exchange–correlation functional is treated by the generalized gradient approximation (GGA)³⁵ with the parametrization scheme of Perdew, Burke, and Ernzerhof (PBE).³⁶ The plane-wave basis set with a cutoff energy of 500 eV is used. A 6×6 hexagonal

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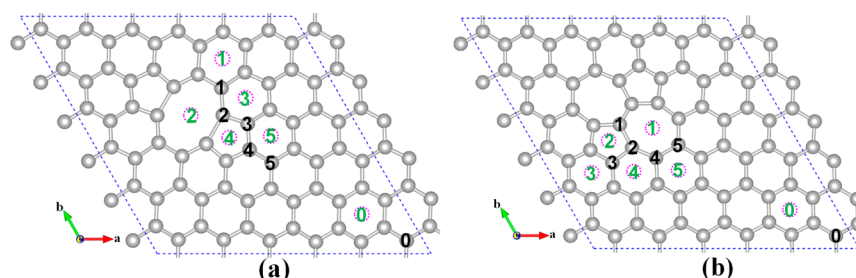


Figure 1. Optimized atomic structures of (a) DV5–8–5 and (b) SW55–77 defect. The hollow sites for Li adsorption are indicated by the dotted circles with numbers.

Table 1. Adsorption Energies [$E_{\text{ad},0}(\text{Li})$, in eV] of Li Adsorbed on the *H*, *B*, and *T* Sites in Perfect Graphene^a

reference	supercell size	$E_{\text{ad},0}(\text{Li})$			$E_{\text{ad},0}(\text{Li}) - E_{\text{ad},0}(\text{Li}_H)$		
		<i>H</i>	<i>B</i>	<i>T</i>	<i>H</i>	<i>B</i>	<i>T</i>
this work	6 × 6	0.315 (−1.290)	0.601 (−1.005)	0.624 (−0.982)	0.000	0.285	0.308
this work	7 × 7	0.298 (−1.308)	0.599 (−1.007)	0.618 (−0.988)	0.000	0.301	0.320
this work	9 × 9	0.190 (−1.416)	0.546 (−1.060)	0.566 (−1.041)	0.000	0.356	0.376
ref 41 ^b	4 × 4	(−1.096)	(−0.773)	(−0.754)	0.000	0.322	0.342
ref 32 ^c	4 × 4	(−1.567)	(−1.240)	(−1.191)	0.000	0.327	0.376

^aValue in parentheses is the adsorption energy of Li given with respect to the total energy of an isolated Li atom. ^bGGA. ^cLocal spin density approximation.

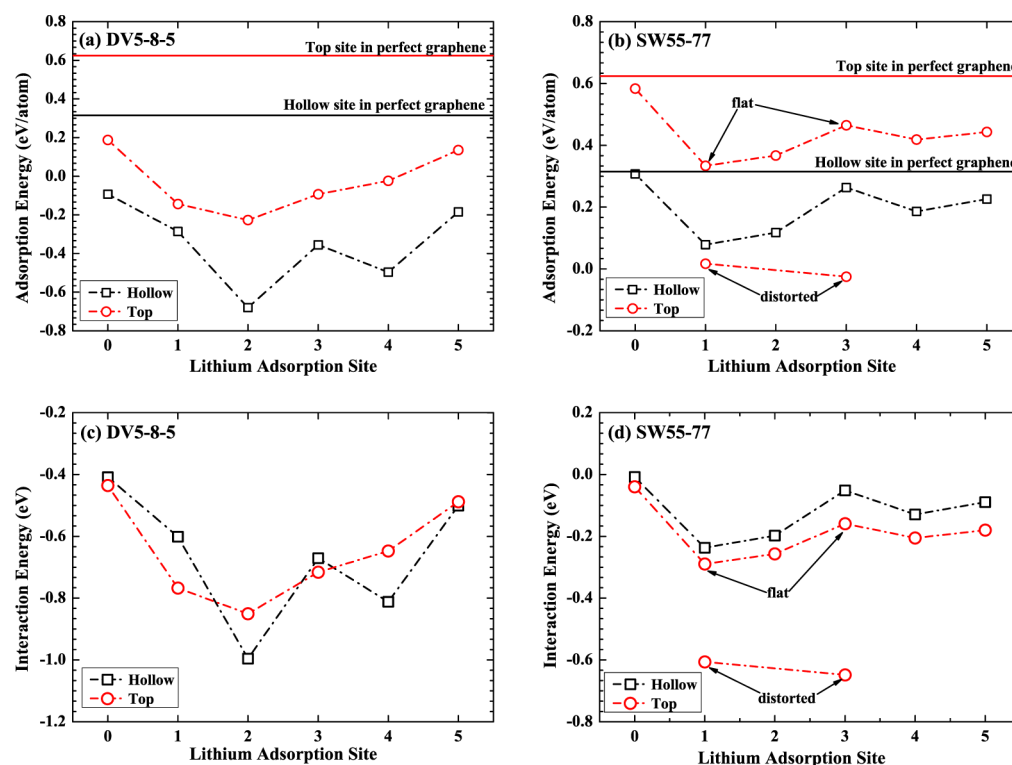


Figure 2. Dependence of the adsorption energy of Li adatom on different sites (shown in Figure 1) in defective graphene with (a) DV5–8–5 and (b) SW55–77 defect. The interaction energy of a Li adatom and a point defect: (c) DV5–8–5 and (d) SW55–77 defect. The adsorption energy of Li adsorbed on the hollow (top) site in perfect graphene is indicated by the black (red) solid horizontal line in panels a and b. In panels b and d, the “flat” means that the *z* coordinates of carbon atoms in graphene sheet are fixed to keep a flat geometry, whereas the “distorted” indicates that the degree of freedom of carbon atoms in graphene sheet is all allowed to relax and that Li adsorption results in a corrugation of graphene sheet.

supercell of graphene is employed to model the point defects and the Li adsorption. The atomic layer of graphene is put in the *xy* plane, the in-plane lattice constant of supercell is 14.81 Å, and a vacuum layer of 12 Å is used in the *z* direction to eliminate the interaction between graphene sheet and its periodic replicas. The Brillouin zone (BZ) integration is

approximated by using a special *k*-point sampling of the Monkhorst–Pack scheme³⁷ with a Γ -centered grid. In the structural relaxation and the static self-consistent-field calculation, a $5 \times 5 \times 1$ *k*-grid is adopted and a $11 \times 11 \times 1$ *k*-grid for the calculations of the density of states (DOS). In the geometry optimization, the atoms are allowed to relax until

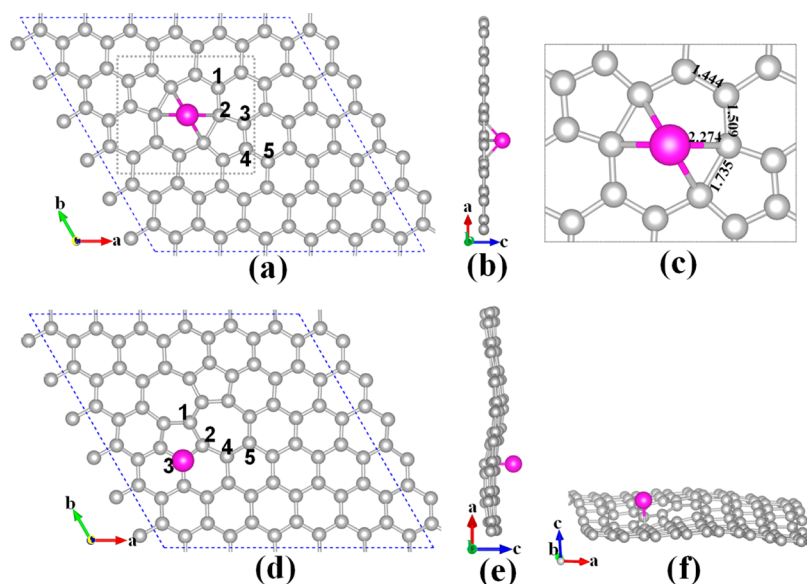


Figure 3. Optimized atomic structure for Li adsorbed on the hollow (H_2 site) of an octagonal ring of DV5-8-5: (a) Top view, (b) side view, and (c) the bond lengths (in angstroms) around Li adatom. The optimized atomic structure for Li adsorbed on the T_3 site of SW55-77. (d) Top view and (e,f) side view.

forces on atoms are <0.02 eV/Å. In the present study, we consider two typical point defects in graphene, that is, divacancy (DV) and Stone–Wales (SW)³⁸ defect. The ordinary configuration of DV is created by removing a C–C dimer from perfect graphene, and the SW defect is formed by an in-plane 90° rotation of a C–C bond in perfect graphene.²³ The optimized atomic structures of DV and SW defect are depicted in Figure 1a,b, respectively. The ordinary configuration of DV contains two pentagons and one octagon (denoted as DV5-8-5), whereas the SW defect has two pentagons and two heptagons (denoted as SW55-77).

For Li adsorbed on perfect graphene, three typical adsorption sites with high symmetry are considered as follows: the hollow (H) site above the center of a hexagon ring, the bridge (B) site above the midpoint of a C–C bond, and the top (T) site directly above a carbon atom. For Li adsorbed on defective graphene, we considered two types of adsorption sites: H site above the center of a polygon (pentagon, hexagon, or octagon) ring and T site. The inequivalent sites (i) in the defect region in addition to site 0 away from the defect for each defective graphene are marked in Figure 1.

To study the effect of point defects on the diffusion of Li on graphene surface, we have calculated the energy barrier ΔE of Li diffusion via ab initio techniques. The climbing image nudged elastic band (CI-NEB) method^{39,40} is used to seek the saddle points and minimum energy path. Four images are employed between two end points. Each image is relaxed until the forces on atom are <0.02 eV/Å.

3. RESULTS AND DISCUSSION

3.1. Energetic Stability of Li Adsorbed on Graphene with Point Defects. We first examine the stability of point defects in graphene by computing their formation energies as follows

$$\Delta E_0(d) = E_d - E_0 + n_C \mu_C \quad (1)$$

where E_0 and E_d are the total energies of perfect graphene and defective one with point defects, respectively. n_C is the number

of the removed carbon atoms and μ_C is the chemical potential of carbon. Here μ_C is taken as the total energy per atom of perfect graphene. For DV5-8-5 and SW55-77 considered here, their formation energies are 7.833 and 5.203 eV, respectively, indicating that SW55-77 is more stable than DV5-8-5. Our results are in good agreement with the previous ones (7.2–7.9 eV for DV5-8-5 and 4.5–5.3 eV for SW55-77, where the variation is caused mainly by the use of different supercell size and the use of different approximation in the treatment of DFT).²³ All of these validate the computational setup chosen in the present study.

To assess the stability of Li adsorbed on graphene with point defects, we calculate the adsorption energy of a Li adatom as follows

$$E_{ad,d}(Li) = E_{Li,d} - E_{Li} - E_d \quad (2)$$

where $E_{Li,d}$ is the total energy of Li adsorbed on graphene with point defects and E_{Li} is taken as the total energy of bulk Li with bcc structure. For the case of Li adsorbed on perfect graphene, the adsorption energy of Li adatom is defined as

$$E_{ad,0}(Li) = E_{Li,0} - E_{Li} - E_0 \quad (3)$$

where $E_{Li,0}$ is the total energy of Li adsorbed on perfect graphene. In literature,^{32,41} $E_{ad,0}(Li)$ is calculated by taking E_{Li} as the total energy of an isolated Li atom. It is evident that the difference in the definition of $E_{ad,0}(Li)$ between the present study and refs 32 and 41 is indeed the cohesive energy $E_{coh}(Li)$ of bulk Li with bcc structure. (The value of $E_{coh}(Li)$ obtained in the present GGA calculations is 1.606 eV/atom, which is in excellent agreement with the experimental result (1.630 eV/atom)⁴².) To describe the interaction between a Li adatom and a point defect, we define an interaction energy as follows

$$E_{int} = (E_{Li,d} + E_0) - (E_{Li,0} + E_d) \quad (4)$$

E_{int} gives the energy difference of the complex of Li adatom and point defect with respect to the sum of the total energies of individual Li adatom and point defect. The negative (positive) sign of E_{int} defined in eq 4 indicates that the formation of a

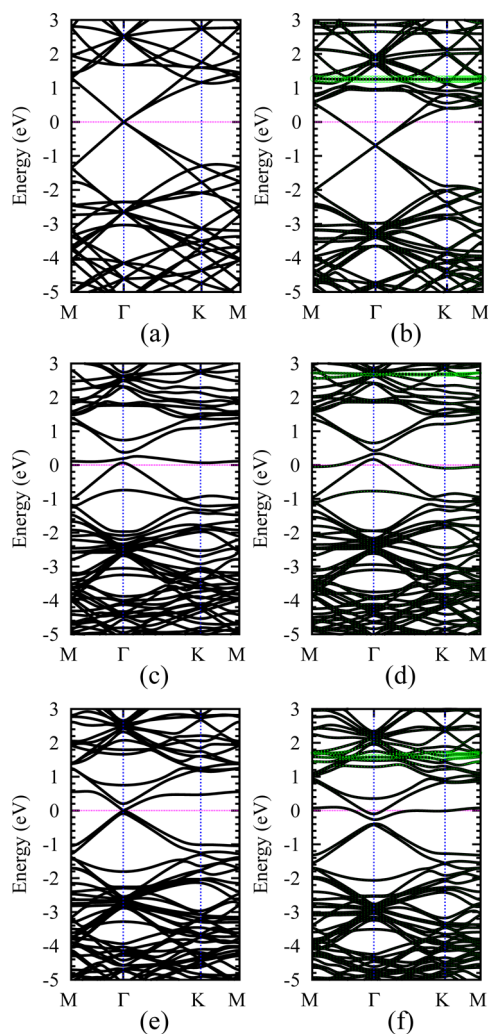


Figure 4. Band structures for graphene before and after absorbing Li: (a) perfect graphene, (b) Li adsorbed on the hollow site (*H*) of a hexagonal ring in perfect graphene, (c) DV5-8-5, (d) Li adsorbed on the hollow site (*H2*) of an octagonal ring of DV5-8-5, (e) SW55-77 defect structure, and (f) Li adsorbed on the *T3* site of SW55-77. The Fermi energy is set to zero. The contributions from Li *s* orbital to the wave function are indicated by the size of green open points in panels b, d, and f.

complex of Li adatom and point defect is energetically favorable (unfavorable). It is obvious that eq 4 can be rewritten in the following way

$$E_{\text{int}} = E_{(\text{Li},\text{d})} - E_{\text{d}} - (E_{\text{Li},0} - E_0) \\ = E_{\text{ad,d}}(\text{Li}) - E_{\text{ad,0}}(\text{Li}) \quad (5)$$

Therefore, the interaction energy of Li adatom and point defect can be regarded as the difference between the adsorption energies of Li adsorbed on defective and perfect graphene.

Table 1 presents the adsorption energy $E_{\text{ad,0}}(\text{Li})$ of Li adsorbed on perfect graphene. The $E_{\text{ad,0}}(\text{Li})$ for *H* site is 0.315 eV, which is lower than those of *B* and *T* sites by 0.285 and 0.308 eV, respectively. Therefore, our results show that the most stable adsorption site of Li adsorbed on perfect graphene is the *H* site, which agrees well with previous studies.^{32,41,43} It should be pointed out that the supercell size (i.e., 6×6) in the present study is larger than the one (i.e., 4×4) in refs 32 and 41. We have checked the supercell size dependence for the Li

adsorption on graphene and find that the use of 6×6 supercell can provide more reliable results. (See Table 1.)

The adsorption energies of Li adsorbed on different sites in graphene with DV5-8-5 and SW55-77 are presented in Figure 2a,b, respectively. In the presence of DV5-8-5 in graphene, the most stable site for Li adsorption is the hollow site of the octagonal ring (i.e., *H2* site as shown in Figure 1a). Li adatom tends to be adsorbed on the hollow site, rather than the top site. This is similar to the situation of Li adsorbed on perfect graphene. For Li adsorbed on different hollow sites above the centers of the polygonal rings, the stability of Li adatom from high to low exhibits the following order: octagonal ring > pentagonal ring > hexagonal ring. It is noted that the adsorption energy of Li on the *H0* site, where the center of the corresponding hexagonal ring is about five C-C bonds away from the one of the octagonal ring of DV5-8-5, is ~ 0.408 eV lower than that of Li adsorbed on the hollow site of perfect graphene, instead of being at the same order. This may be caused by the following reason: the defect states induced by DV5-8-5 are quite extended along the direction of C4-C5 bond, as revealed by the STM images,²⁹ and the *H0* site considered here is not far enough away from the vacancy site. Compared with Li adsorbed on the *H0* site, the adsorption energies for Li adsorbed on the hollow sites around defect region become more negative (i.e., corresponding to an exothermic process), indicating that the presence of DV5-8-5 can enhance the Li adsorption. This also can be seen from the interaction energy of a Li adatom and a DV5-8-5, as shown in Figure 2c. The values of E_{int} for Li adsorbed on the hollow sites are negative, indicating that Li adatom and DV5-8-5 attract each other.

Figure 3 shows the atomic structure for the most stable site (*H2*) of Li adsorbed on graphene with DV5-8-5. Meanwhile, the distance ($d_{\text{Li-C}}$) between the adsorbed Li atom and its nearest carbon atoms and the C-C bond lengths around Li adatom are also given in Figure 3c. The Li-C bond length for Li adsorbed on the hollow site (*H2*) above the center of the octagonal ring of DV5-8-5 is 2.274 Å, which is slightly larger than that for Li adsorbed on the hollow site in perfect graphene by 0.011 Å. The height of Li adsorbed on *H2* site in DV5-8-5 is 1.404 Å, which is ~ 0.03 Å less than the one in the case of perfect graphene.

For graphene with SW55-77, Li adsorbed on the hollow site of a heptagonal ring is more stable than other hollow sites of pentagonal and hexagonal rings. Similar to the case of DV5-8-5, Li adatom energetically prefers the sites around the defect region to the ones in bulk region. The interaction energy of a Li adatom and a SW55-77 is presented in Figure 2d, and it can be seen that the values of E_{int} for Li adsorbed on the sites near the defect region are negative, indicating an attractive interaction for Li adatom and SW55-77. However the interaction of Li adatom and SW55-77 is weaker than that of Li adatom and DV5-8-5. The *T3* site (see Figure 1b) is the most stable one for Li adsorption. It should be pointed out that the Li adsorption on *T3* site induces a significant structural distortion of graphene, as shown in Figure 3d-f. In this case, the graphene sheet does not keep flat, and it is distorted into a sinelike buckled one,⁴⁴ where the atoms at the defect core (i.e., the rotated C1-C1 bond) move out of plane in opposite directions. The buckling height (i.e., the difference along the *z* axis between the highest and lowest carbon atom in the supercell)⁴⁴ is ~ 1.358 Å, which is slightly larger than that (i.e., 1.198 Å in the present calculation with a 6×6 supercell or 1.16

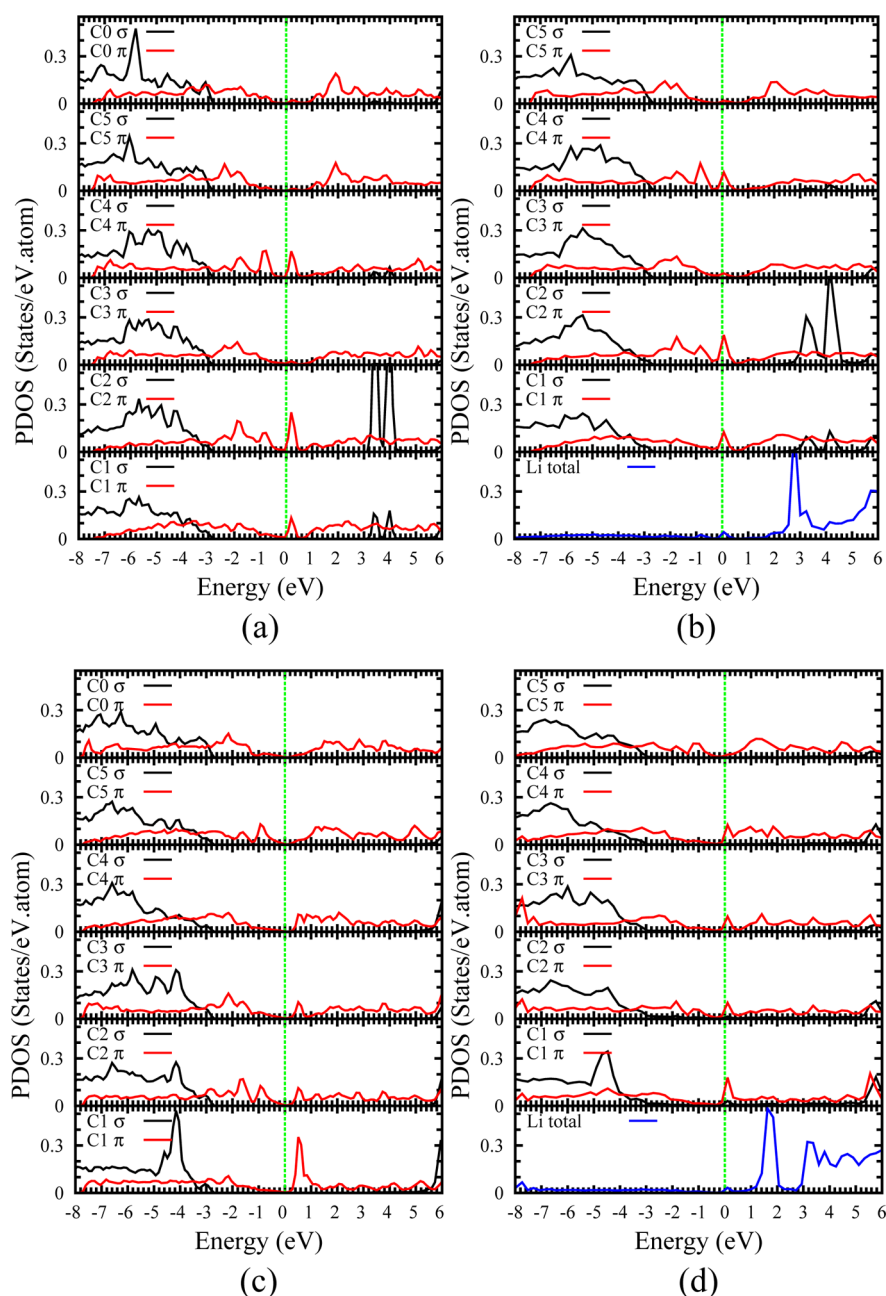


Figure 5. Partial density of states (PDOS) for atoms around defect region in defective graphene: (a) DV5–8–5, (b) Li adsorbed on the H2 site of DV5–8–5, (c) SW55–77, and (d) Li adsorbed on the T3 site of SW55–77. The Fermi energy is set to zero.

Å reported in ref 44 with a 5×5 supercell) of sinelike buckled SW55–77 without Li adsorption. We find that the adsorption of Li on the T1 site also results in a structural distortion of graphene sheet, in which the flat SW55–77 structure is buckled into a cosinelike one (i.e., the C1 atoms at the defect core move out of plane along the z axis in the same direction)⁴⁴ and the buckling height is ~ 1.118 Å (0.984 Å for cosinelike buckled structure without Li adsorption). As pointed out by Ma et al. in ref 44 according to the vibrational analysis, the flat SW55–77 defect structure has two imaginary frequencies, which lead to two buckled (i.e., out-of-plane) wavelike defect structures (i.e., the sinelike and cosinelike buckled SW55–77), the sinelike buckled structure is a true minimum with no imaginary frequencies, whereas the cosinelike structure is a transition state with one imaginary frequency. The higher stability of sinelike

buckled SW55–77 structure than the flat one is ascribed to the strain relief achieved through the out-of-plane displacements.⁴⁴ In the flat SW55–77 structure, the rotated C1–C1 bond is compressed to 1.315 Å from the equilibrium C–C bond length of 1.42 Å, whereas the C1–C1 bond length in the sinelike buckled structure is ~ 0.02 Å longer than that in the flat one. After Li adsorption on the T3 site, the C1–C1 bond length is elongated to 1.413 Å compared with the flat SW55–77 structure, indicating a more significant release of the compression. For the adsorption of Li on the T1 and T3 sites of SW55–77, we have also checked the “flat” cases by fixing the z coordinates of all carbon atoms in the same plane during the geometry optimization. As presented in Figure 2c, the buckling of SW55–77 defect structure induced by the adsorption of Li on the T1 and T3 sites lowers the adsorption

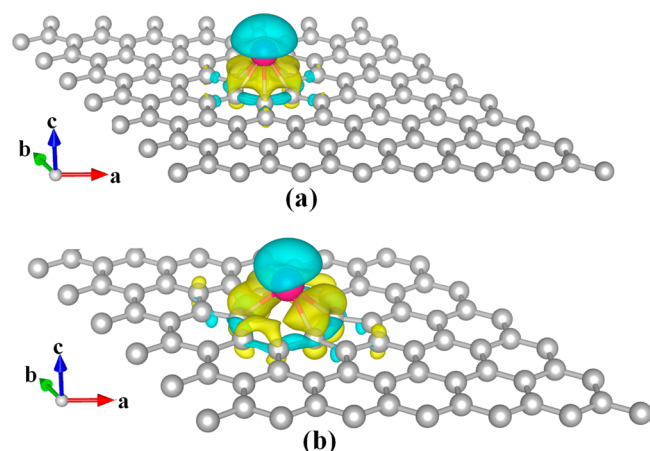


Figure 6. Isosurface ($2 \times 10^{-3} \text{ e}/\text{\AA}^3$, yellow for $\Delta\rho > 0$ and blue for $\Delta\rho < 0$) of the difference charge density $\Delta\rho$ for the most stable configuration of Li adsorbed on (a) perfect graphene and (b) defective one with DV5-8-5.

energy by about 0.316 and 0.489 eV, respectively. Without Li adsorption, the cosinelike and sinelike buckled SW55-77 defect structures are lower than the flat one by 0.077 and 0.232 eV (0.269 eV reported in ref 44 using a 5×5 supercell) in the present calculation with a 6×6 supercell, respectively. For other adsorption sites of the flat SW55-77 defect structure, the Li adsorption does not induce a buckling distortion.

3.2. Electronic Structures of Graphene Modified by Li Adatom. In this subsection, we will discuss how the electronic structure of graphene is modified by point defects and the adsorbed Li atom.

As a reference, we first present the band structures of perfect graphene without and with Li adsorbed on the hollow site in Figure 4a,b, respectively. For perfect graphene, the original K and K' points in the first BZ of primitive cell are folded into the BZ center (i.e., Γ point) of 6×6 super cell, which is used in our calculations. It is known that for perfect graphene, the Fermi level (E_F) coincides with the Dirac point. After adsorbing Li on the hollow site of graphene, E_F obviously shifts up into the conduction band due to the electron donation from the Li adatom to the π^* bands of graphene, whereas the 2s states of Li adatom lie about 1.3 eV above E_F due to the charge transfer.

Figure 4c shows the band structure of graphene with DV5-8-5. It can be seen that there is a nearly flat band appearing just above the Fermi level, which is consistent with recent experimental study and the combined DFT calculations.²⁹ From the partial density of states (PDOS) for C atoms around the defect region depicted in Figure 5a, it is found that the main contributions of such flat band are dominated by the p_z states of C2 and C1 atoms next to vacancy site and C4 atoms. This is consistent with the previous studies⁴⁵⁻⁴⁸ of tight-binding model, which predicted that a single impurity in sublattice A of graphene induces an impurity state mostly localized in

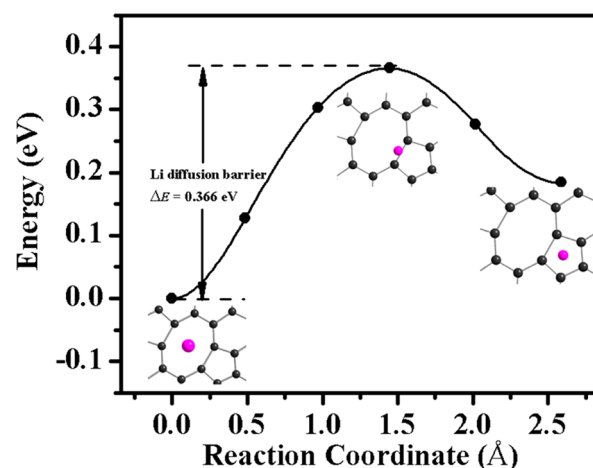


Figure 7. Energy barrier ΔE along the Li migration path of $H2 \rightarrow H4$ on graphene with DV5-8-5. The diffusion barrier ΔE is defined as the energy difference between the saddle point and the local potential energy minimum corresponding to the initial state.

sublattice B and vice versa due to the existence of two nonequivalent Dirac points. Because C3 and C5 atoms belong to a sublattice different to the one of C2 and C4 atoms, there is almost no contribution from the former two atoms to the defect states just above the Fermi level. Figure 4d shows the band structure of graphene with Li adsorbed on the H2 site of DV5-8-5. The nearly flat defect band of DV5-8-5 is pushed down, and it becomes partially occupied due to the electron donated by Li adatom; however, the Fermi level is still below the Dirac point, indicating a p-type doping. This is due to the interference of the defect state of DV5-8-5, resulting in the behavior of Li adatom different to that of Li adsorbed on perfect graphene. In the latter case, Li adsorption introduces electron carriers to graphene. The PDOS of Li and C atoms around defect region in DV5-8-5 are shown in Figure 5b. It can be seen that the defect p_z states localized on C2 and C4 atoms are significantly affected by Li adsorption and become partially occupied, whereas the states for C3 and C5 atoms are almost unchanged because of no contribution from them to the defect states of DV5-8-5.

Figure 4e shows the band structure of graphene with SW55-77 defect. The Fermi level still coincides with the Dirac point; however, the degeneracy of the π^* bands at the Dirac point is removed due to the in-plane 90° rotation of a C-C bond. A nearly flat band appears at ~ 0.6 eV above E_F , and it mainly consists of the p_z states of the atoms (i.e., C1) at the defect core of SW55-77. (See Figure 5c.) This defect state also shows a small tail on the C3 and C4 atoms because they belong to a same sublattice of C1 atom. Figure 4f shows the band structure of graphene with Li adsorbed on the T3 site of SW55-77 defect. As discussed above, in this case, the SW55-77 defect is sinelike buckled. We can see that the Fermi level shifts up into the conduction band and becomes above the Dirac point,

Table 2. Energy Barrier (ΔE , in eV) for Li Diffusion on the Graphene Surface with DV5-8-5 and SW55-77 Defects^a

	defect free	DV5-8-5				SW55-77
		H2 → H1	H2 → H3	H2 → H4	H4 → H5	H1 → H5
ΔE	0.277	0.538 (0.140)	0.455 (0.122)	0.366 (0.182)	0.443 (0.117)	0.370 (0.236)

^aResult for Li diffusion from a hollow site to its nearest neighbor one on the defect-free graphene is also listed for comparison. The values in parentheses correspond to the ΔE for the backward diffusion of Li atom.

indicating an *n*-type doping. The electron donation of Li adatom also pushes down the nearly flat defect band of SW55–77 defect, and this band becomes partially occupied. In the case of Li adsorbed on the T3 site of SW55–77 defect, the sinelike buckling leads to a release of the compressed strain, and the C1–C1 bond length is elongated by 0.098 Å with respect to the flat SW55–77 defect structure, as discussed above. Therefore, the contribution of C1 atoms to the defect band of SW55–77 is reduced significantly after Li adsorption on the T3 site. (See Figure Sd.) Because of the charge transfer, the 2s states of Li adatom lie at ~1.7 eV above E_F .

Charge transfer is most sensible to ionic bonding and less relevant to covalent.⁴¹ To understand the bonding nature of Li adsorbed on graphene, we calculate the difference charge density $\Delta\rho$, which is defined as follows

$$\Delta\rho = \rho_{\text{Li+G}} - [\rho_{\text{G}} + \rho_{\text{Li}}] \quad (6)$$

where $\rho_{\text{Li+G}}$ represents the charge density for the total system (Li and graphene), ρ_{G} is the charge density of a graphene (without Li), and ρ_{Li} is the charge density of an isolated Li atom located at the same position as in the total system. Figure 6a,b shows the difference charge densities of Li adsorbed on perfect graphene and the defective one with DV5–8–5, respectively. In both cases, there is a net gain of electronic charge in the intermediate region between Li and graphene, whereas there is a net loss of electronic charge just above the Li, indicating a significant charge transfer from the adsorbed Li to its nearest neighbor C atoms. In an effort to quantitatively estimate the amount of charge transfer between the adsorbed Li and the graphene substrate, we have performed the Bader charge analysis.^{49,50} The Bader charge state of Li atom adsorbed on the H2 site of DV5–8–5 is about +0.907|e|, which is nearly the same as in the defect-free graphene (+0.904|e| for Li). The averaged Bader charge state of four carbon atoms next to the vacancy site is –0.195|e|. All of these suggest that the interaction between the adsorbed Li atom and its nearest neighbor C atoms is predominantly ionic and that valence electrons of the adsorbed Li atoms are transferred to the nearest neighbor C atoms.

3.3. Diffusion of Li Adatom on Graphene. Because Li atom tends to be adsorbed on the hollow site of graphene, we consider only the case for Li diffusion from the most stable adsorption position to the neighboring hollow sites. For Li diffusion on defective graphene with DV5–8–5, we consider three cases, that is, the diffusion of Li adatom from H2 site to H1, H3, and H4 ones (H2 → H1, H2 → H3, and H2 → H4, respectively), respectively. In the case of SW55–77, one path of Li diffusion is studied as an example, that is, Li diffusion from H1 site to H5 one (H1 → H5). The calculated energy barriers (ΔE) for these diffusion paths of a Li adatom are summarized in Table 2. The diffusion path of H2 → H4 is studied in detail as a representative, and the calculated ΔE and transition state are presented in Figure 7. The results for other diffusion paths of Li adatom around DV5–8–5 and SW55–77 defects are presented in Figures S1–S4 in the Supporting Information. Compared with the Li diffusion from a hollow site to a nearest neighboring one on perfect graphene (0.277 eV for ΔE , see Figure S6 in the Supporting Information), we can see that the ΔE for Li migration from H2 to H4 increases by 0.089 eV and the one for the corresponding backward diffusion decreases by 0.092 eV. The similar trends can also be found in other diffusion paths of H2 → H1 and H2 → H3. These suggest that the Li adatom tends to diffuse toward the DV site, and it is

unfavorable to diffuse outward the DV site.⁴³ Therefore, the Li adatom may be trapped at the hollow site above the octagonal ring of DV5–8–5. In the case of SW55–77 defect, it is found that the Li adatom diffuses easily from the hollow sites above the hexagonal rings to the one above the heptagonal ring. Therefore, the presence of DV5–8–5 and SW55–77 defects would trap the Li adatom, which is similar to the zigzag edge of graphene nanoribbons.³²

4. CONCLUSIONS

In conclusion, we have performed the first-principles calculations to study the Li adsorption and diffusion on graphene with DV and SW defect. Our results show that Li adatom prefers the hollow site above the center of octagonal (heptagonal) ring to the one of pentagonal (hexagonal) ring if they exist in defective graphene. The interaction of a Li adatom and a DV is attractive, and it is stronger than that of a Li adatom and a SW defect. The adsorption of Li on the top sites of carbon around SW defect would result in a buckling of graphene sheet. Our results suggest that the intentional creation of point defects before absorbing Li would enhance the Li adsorption significantly.

■ ASSOCIATED CONTENT

Supporting Information

Energy barriers along the Li migration paths of H2 → H1, H2 → H3, and H4 → H5 on divacancy, the energy barrier for Li migration path of H1 → H5 on Stone–Wales defect, and the energy barrier for Li diffusion between two nearest neighboring hollow sites on perfect graphene. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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